

SAMPLE 1 MID TERM 1

- To address 1-MB of memory we need 20 bits of addressing, then how much megabytes of memory can be accessed using 30 bits of addressing. _____ MB.
- Write an instruction which reads the first byte of its own op-code and stores it into al register.
- Given the following sequential set of instructions of same program, write down the values of CF, PF, ZF after each instruction: (initially all flags are zero.)

	CF	PF	ZF
xor ah, ah			
mov al, 0x4A			
shl al, 2			
rcr ah, 3			
sub ah, al			

- Write assembly code to compare two 32-bit numbers such that if num1 is equal to equal to num2 it sets ZF=1 else ZF=0. (Declare two 32-bit numbers in memory to compare)
- Write an assembly program, such that given an array of **ten** integers (each integer is stored as word), your program finds and stores the sum of unique elements of array in a memory label called **"sum"** (defined word).
For Example is array is 3,3,4,5,1,1
Sum should be 3+4+5+1=13
- In the following code, tell whether the jump will be taken or not. **Show each and every step** to get marks.

```
[org 0x100]
mov ax, 0xF193
mov bx, 0x1234
xor ax, 0x195B
cmp ax, bx
jl exit
exit:
mov ax, 0x4c00
int 21h
```

SAMPLE 2 MID TERM 1

- What changes will be made to 'arr' after the program execution? Fill the table below:

<pre> [org 0x100] 1. mov si, -1 2. mov di, [size] 3. 4. l1: 5. add si, 1 6. cmp si, di 7. je end 8. mov al, [arr+si] 9. shr al, 1 10. jnc l1 11. 12. l2: 13. sub di, 1 14. cmp si, di 15. je end 16. mov al, [arr+di] 17. shr al, 1 18. jc l2 </pre>	<pre> 19. cmp si, di 20. jnl end 21. mov al, [arr+si] 22. mov ah, [arr+di] 23. mov [arr+di], al 24. mov [arr+si], ah 25. jmp l1 26. end: 27. 28. mov ax, 0x4c00 29. int 21h 30. arr: db 3, 10, 2, 0, 7, 5, 8 31. size: dw 7 </pre>
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- Rewrite the above code from line 4 to 10 without using shr.
- Fill the following table. These instructions are from same program and are not independent. Write the corresponding output for the given registers' and flags' values. [10 Marks] AX=0x5CAA DX=0x3729 CX=0x235A

Instructions	Updated value after executing the instruction			Flag values after the instruction execution		
	AL	DL	CL	CF	OF	SF
xor al, dl						
add dl, dl						
sub cl, dl						
sar al, cl						
adc al, dl						

- Implement 32 Bit Subtraction and write answer back into memory. [10 Marks]

Assume that your Architecture only supports these instructions.

Mov, shl, shr, rcl, ror, rol, add, and, not

Note: No Credit would be given if any other instruction is used.

SAMPLE 3 MID TERM 1

- What will be the value of ax in **hexadecimal** after the execution of the following piece of code? Show your working to get full marks.

```
[org 0x100]
mov al,[num1]
mov ah, [num2]
add al, ah

num1: dw 0x01F3
num2: dw 0x1D00
```

- Following is a portion of listing file for the code given in **Question 1**.

```
[org 0x0100]
00000000 A0[0E00]    mov al, [num1]
00000003 8A26[1000]  mov ah,[num2]
00000007 00E0      add al, ah
```

- (i) What is the IP address of the instruction **add al, ah**?
- (ii) Given CS = 0x1234 and DS = 0x3456, what is the physical address of the location where **num1** is placed?
- Look at the following code carefully and write in one line what is the code doing. Your answer has to be 100% correct to gain full marks.

```
[org 0x0100]
movbx, 63
shr byte[number+bx], 1

l1: decbx
rcr byte[number+bx], 1
cmpbx, 0
jne l1

mov ax, 0x4c00
int 21h
; assumenumber is a data label with 64 bytes integer defined.
```

- Write an assembly program that checks **in binary** whether a 16-bit number is palindrome or not. Move 1 in dx register if it is a palindrome else move 0 in dx register. Palindrome is a number which reads the same backward or forward.

For example

0xA425 is a palindrome.

